

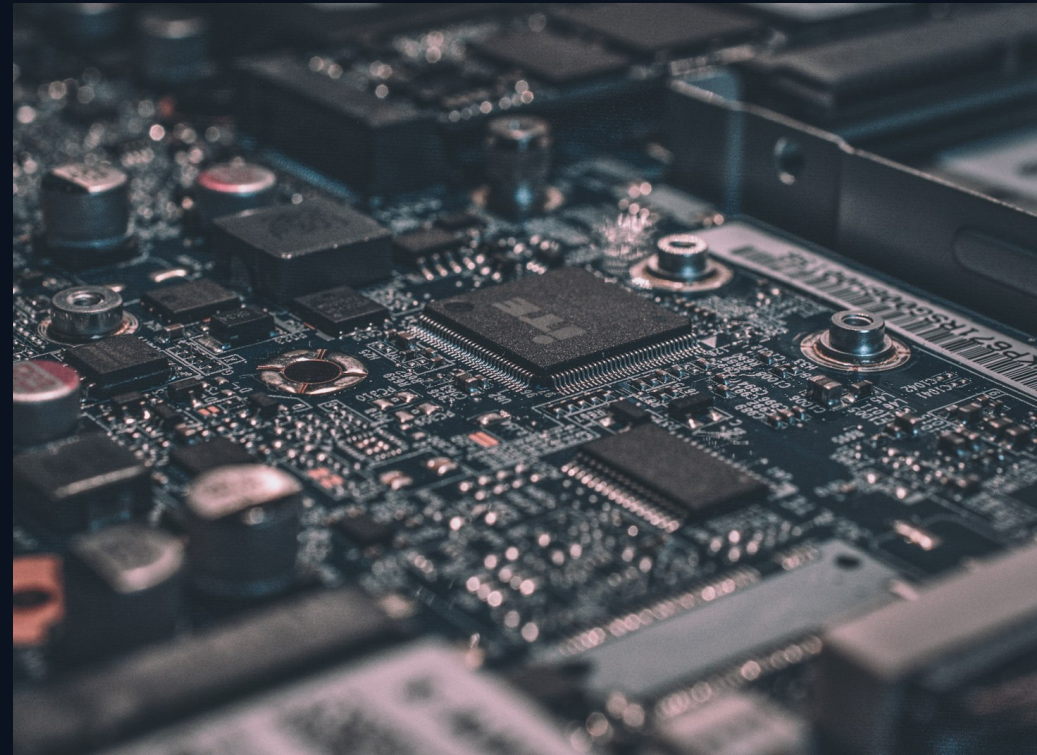


AegisAI

Agentic Prompt Injection Firewall

A runtime security layer between untrusted content and agent tool actions.

ET AI Hackathon 2026 — Agentic Edition
Problem 2 · Pitch Deck



Agents read more than chat.

AI agents ingest documents, websites, emails, and tool outputs.

Hidden instructions can try to:

- Override system intent (prompt injection)
- Hijack tool calls (email, delete, shell)
- Steal secrets or credentials
- Poison MCP tool definitions

Detection alone is not enough — a compromised plan must still be stopped before execution.



AegisAI sits between untrusted content and agent actions.

Runtime security layer — the frontend never decides ALLOW / BLOCK.

01

DETECT

Rules + Prompt Guard +
Safeguard

02

UNDERSTAND

Evidence fusion & risk scoring

03

DECIDE

ALLOW · REVIEW · BLOCK

04

PROTECT

Authz + Tool Firewall + MCP

05

AUDIT

Hashed events, no raw prompts

Defense in depth — detect, then stop the tool boundary.

Detection pipeline

1. Untrusted input → normalization
2. Deterministic rules (9 detectors)
3. Groq Prompt Guard + Safeguard
4. Fusion → Risk → Policy

5. ALLOW | REVIEW | BLOCK

Provider failure → UNCERTAIN

Never silent BENIGN

Protect pipeline

1. Agent security workflow
2. Authn → Authz / tenant
3. Tool Firewall
4. Approval + replay protection
5. Mock MCP gateway
6. Audit → Dashboard

Unauthorized tool actions still stop.

DEMO HIGHLIGHTS

1 Benign request

ALLOW / REVIEW — safe path

2 Direct injection

BLOCK / REVIEW — tools not executed

3 Indirect injection

Untrusted document content contained

4 Intent hijack

Tool Firewall DENY

5 High-risk delete

REQUIRES_APPROVAL (bound)

6 MCP tamper / shadow

Integrity DENY — mock MCP only

EVIDENCE

0.9706 Live detection F1 P 0.9925 · R 0.9496	45/45 Security invariants INV-01 ... INV-45	11/11 Authorization eval Phase 16 matrix
15/15 MCP eval Mock gateway only	371 Backend tests pytest suite	Hash-only Audit trail No raw prompts / tokens

Detecting injection text is necessary — not sufficient.

- Fail-closed on auth, unknown tools, tenant mismatch, replay
- Uncertainty → REVIEW, never silent BENIGN
- Policy BLOCK cannot be overridden by approval
- Tool Firewall is the final execution boundary
- MCP / tool outputs remain untrusted
- Conservative REVIEW volume is intentional



Protect the tool boundary — even when the model is confused.

AegisAI is a hackathon prototype, not a production security product.
Mock MCP and mock tools keep the demo safe while proving the architecture.

github.com/VarunPunzuri99/AegisAI

Detect → Understand → Decide → Protect → Audit

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